



THE MIND'S EAR

A great composer, Ludwig van Beethoven (1770-1827), noticed deafness approaching before his 30th birthday. He resolved to defy his handicap and composed some of his finest works by hearing the notes in his head.

Ears and hearing

AFTER SIGHT, HEARING IS THE SENSE that provides the brain with most information about the outside world. Compared with other animals, the human ear registers a fairly wide range of sounds. These vary in volume from the delicate notes of a flute to the ear-splitting chords of a heavy metal electric guitar. They vary in pitch from the deep thunderous roar of a jet engine to the high trills of bird song. Ears did not figure greatly in the writings of ancients such as Aristotle of Greece and Galen of Rome. The Egyptians knew that ears hear, but they also thought that ears are involved in breathing.

The scientific study of hearing began in earnest in the 1500s. *The Examination of the Organ of Hearing*, published in 1562, was probably the first major work devoted to ears. Its author was Bartolomeo Eustachio (1520-1574). His name lives on in the eustachian tube, or the pharyngo-tympanic tube, which connects the air-filled middle ear cavity to the back of the throat.

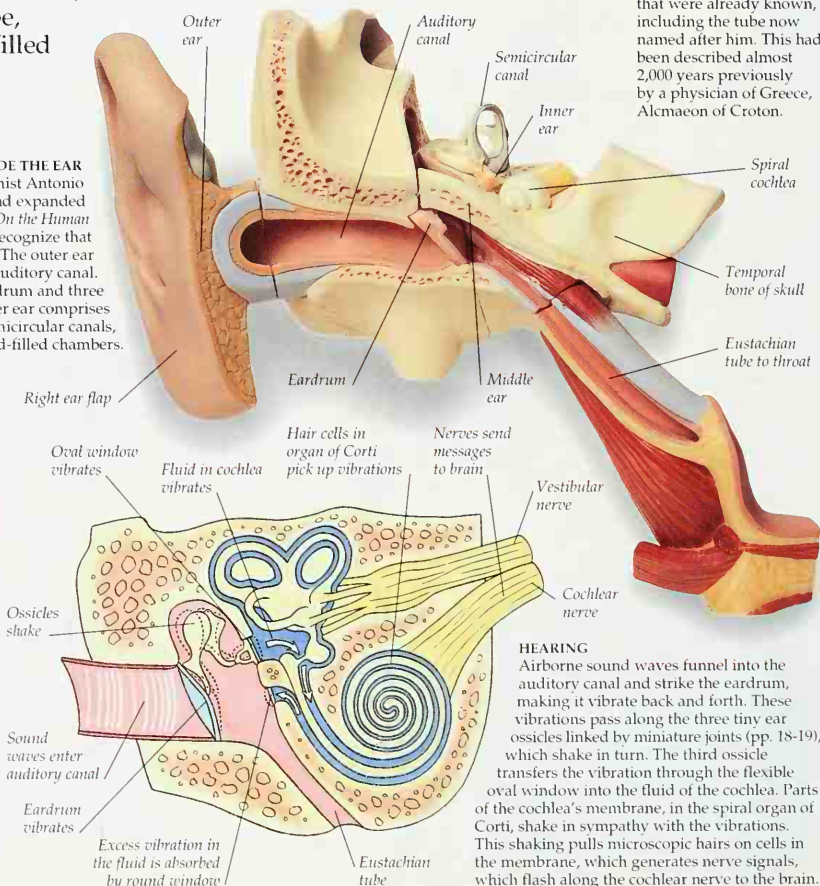
INSIDE THE EAR

Italian mini- and microanatomist Antonio Valsalva (1666-1723) updated and expanded Eustachio's book with his own *On the Human Ear* in 1704. He was the first to recognize that the ear has three main parts. The outer ear consists of the ear flap and the auditory canal. The middle ear contains the eardrum and three tiny bones, the ossicles. The inner ear comprises the snail-shaped cochlea, semicircular canals, and other fluid-filled chambers.



THE EARDRUM

This taut membrane, like stretched skin, is slightly smaller than a little finger's nail. It marks the divide between the outer ear and the middle ear. It can be seen directly or photographed by placing a medical instrument, the otoscope, into the auditory canal. Through the eardrum is a hazy view of the first of the ear ossicles, the hammer. The "handle" of the hammer is firmly attached across half of the eardrum.



HEARING

Airborne sound waves funnel into the auditory canal and strike the eardrum, making it vibrate back and forth. These vibrations pass along the three tiny ear ossicles linked by miniature joints (pp. 18-19), which shake in turn. The third ossicle transfers the vibration through the flexible oval window into the fluid of the cochlea. Parts of the cochlea's membrane, in the spiral organ of Corti, shake in sympathy with the vibrations. This shaking pulls microscopic hairs on cells in the membrane, which generates nerve signals, which flash along the cochlear nerve to the brain.



ECLIPSED BY VESALIUS
Eustachio studied as a professor in Rome at about the same time that Andreas Vesalius was revolutionizing anatomy in Padua. His book on the ear covers several parts that were already known, including the tube now named after him. This had been described almost 2,000 years previously by a physician of Greece, Alcmaeon of Croton.